

Chapter 4 Network Layer: Data Plane

Network layer: “data plane” roadmap

Data plane: 如何從路由器中的一個端口到另一個端口

Control plane : 決定資料如何經過各式路由器

Two key network-layer functions

forwarding: move packets from a router’s input link to appropriate router output link

routing: determine route taken by packets from source to destination

Network layer: data plane, control plane

	位置	解釋
Data Plane	local	Routing data from port to port in router
Control Plane	network-wide	Routing data from router to router in network

Two control-plane approaches

Per-router control plane (Traditional)

→ Implement routing table and algorithms in each router, and send data refer to

Software-Defined Networking (SDN) control plane

→ Remote controller computes, installs forwarding tables in routers

e.g. 有一個上帝在綜合網路清況進行路由判斷

Network service model

Longest prefix matching

Use longest address prefix that matches destination address

Destination Address Range	Link interface
11001000 00010111 00010*** *****	0
11001000 00010111 00011000 *****	1
11001000 00010111 00011*** *****	2
otherwise	3

11001000 00010111 00010110 10100001 → PORT 0

11001000 00010111 00011000 10101010 → PORT 1

Often performed using ternary content addressable memories (TCAMs)

content addressable: present address to TCAM: retrieve address in

one clock cycle, regardless of table size

Cisco Catalyst: ~1M routing table entries in TCAM

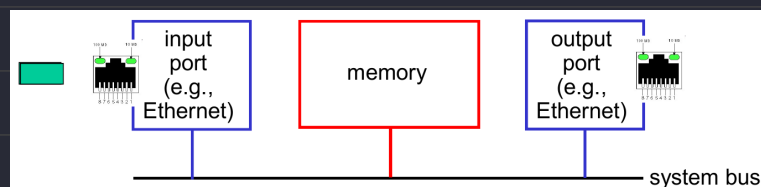
Switching fabrics

Switching rate: Switching rate N times line rate desirable with N inputs

(如果每個 PORT 同時進入一個 PACKET，同時能處理完N個封包就不用 Queue)

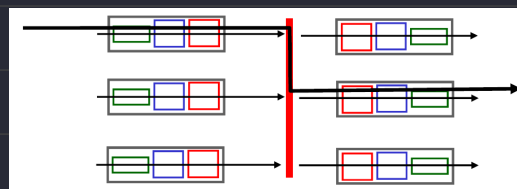
Types of switching fabrics:

(1) Memory



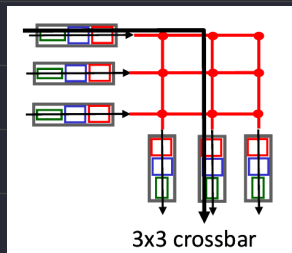
Packet copied to system's memory, and speed limited by memory bandwidth

(2) Bus Contention



Switching speed limited by bus bandwidth

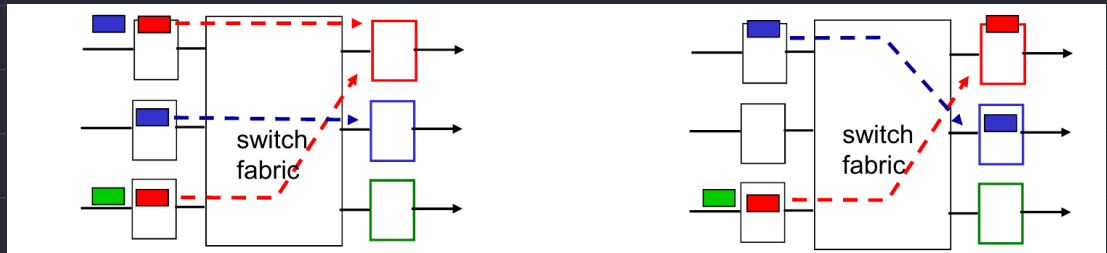
(3) Interconnection network



TODO

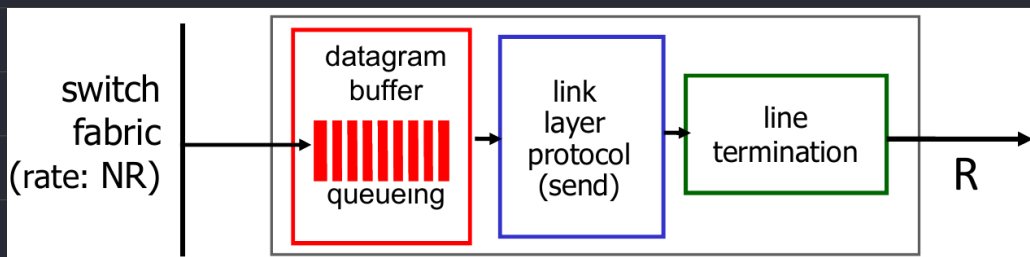
Input port queuing

Head-of-the-Line (HOL) blocking



Green packet is blocked by red packet, it's experiencing HOL blocking

Output port queuing



Because of input speed is higher than output speed

Drop policy: Which packet to drop if no free buffer ?

Datagrams can be lost due to congestion, lack of buffers

Scheduling discipline: Drop packet when buffer is **almost** full

Priority scheduling – who gets best performance, network neutrality

How much buffering?

But too much buffering can increase delays

Buffer Management

drop: which packet to add, drop when buffers are full

- tail drop: drop arriving packet
- priority: drop/remove on priority basis

marking

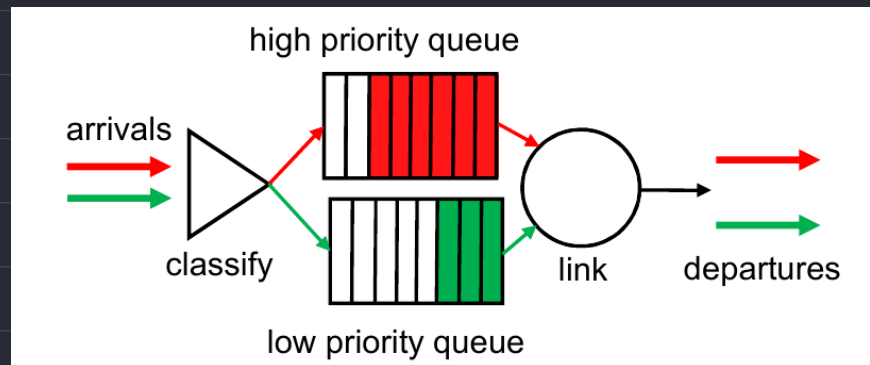
which packets to mark to signal congestion (ECN, RED)

Packet Scheduling: FCFS (First come first serve)

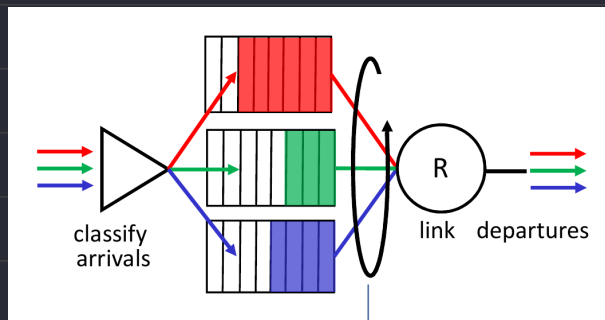
Packet scheduling: deciding which packet to send next on link

First come, first served

→ Like First In First out



Round Robin (RR) scheduling:



輪流送資料

Round Robin Advanced scheduling:

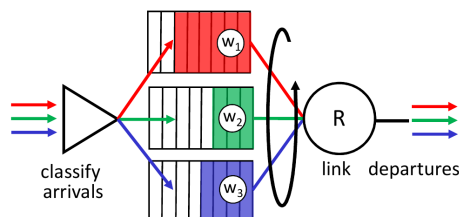
正常 RR + 權重，然後重要的送多次，其他的送一次

Weighted Fair Queuing (WFQ):

- generalized Round Robin
- each class, i , has weight, w_i , and gets weighted amount of service in each cycle:

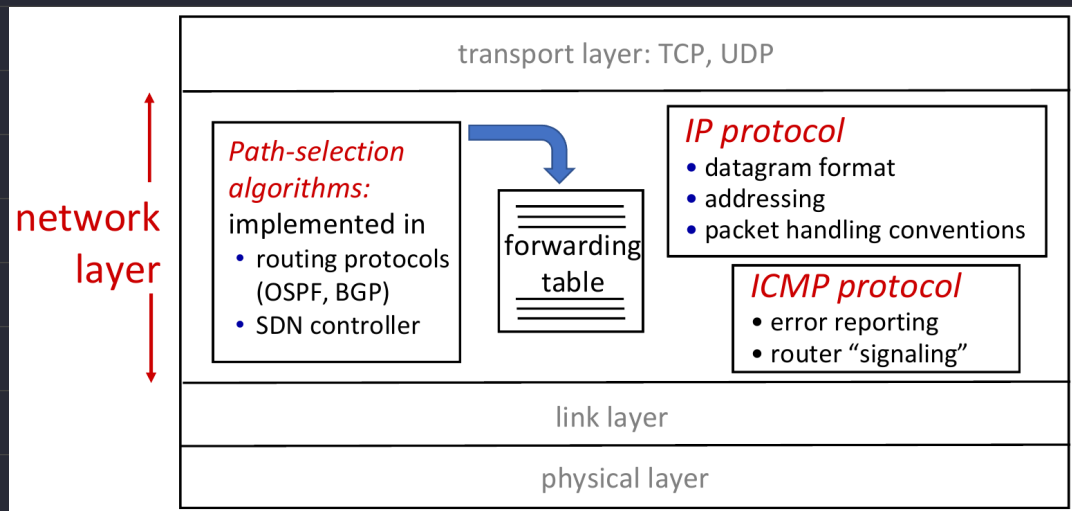
$$\frac{w_i}{\sum_j w_j}$$

- minimum bandwidth guarantee (per-traffic-class)



IP: the Internet Protocol

Network Layer: Internet



ver	head. len	type of service	length
16-bit identifier	flags	fragment offset	
time to live	upper layer	header checksum	
source IP address			
destination IP address			
options (if any)			
payload data (variable length, typically a TCP or UDP segment)			

(32-bit source IP address)

(32-bit destination IP address)

(timestamp, record route taken)

Ver: The version of the IP portocol

Header len (bytes): Represent the size of the header section (IP 標頭)

"Type" of service: diffserv, ECN

Length (bytes): total datagram length

16-bit identifier, flags, fragment offset:

fragmentation / reassembly (如果封包太大無法被一次送出, 路由器會切封包, metadata放這)

Time to live: 註明封包最多可以經過多少 Router, 每經過一個 Router 就會 -1

Header check sum: 使用 1 補數糾錯, 經過一個 Router 會重新計算變更後的 checksum

→ Basic overhead = 20 (TCP) + 20 (IP) = 40 Bytes

→ + app layer overhead for TCP + IP

Subnets

Definition:

Device interfaces that can physically reach each other without passing through an intervening router

Subnet part:

Devices in same subnet have common high order bits

Host part:

Remaining low order bits

CIDR: Classless InterDomain Routing

192.168.1.1/24 : 24 means that the first 24bits is subnet: 192.168.1.XXX

DHCP: Dynamic Host Configuration Protocol

Problems

1. Q: How does a host get IP address within its network (host part of address)
2. Q: How does a network get IP address for itself (network part of address)

Goals:

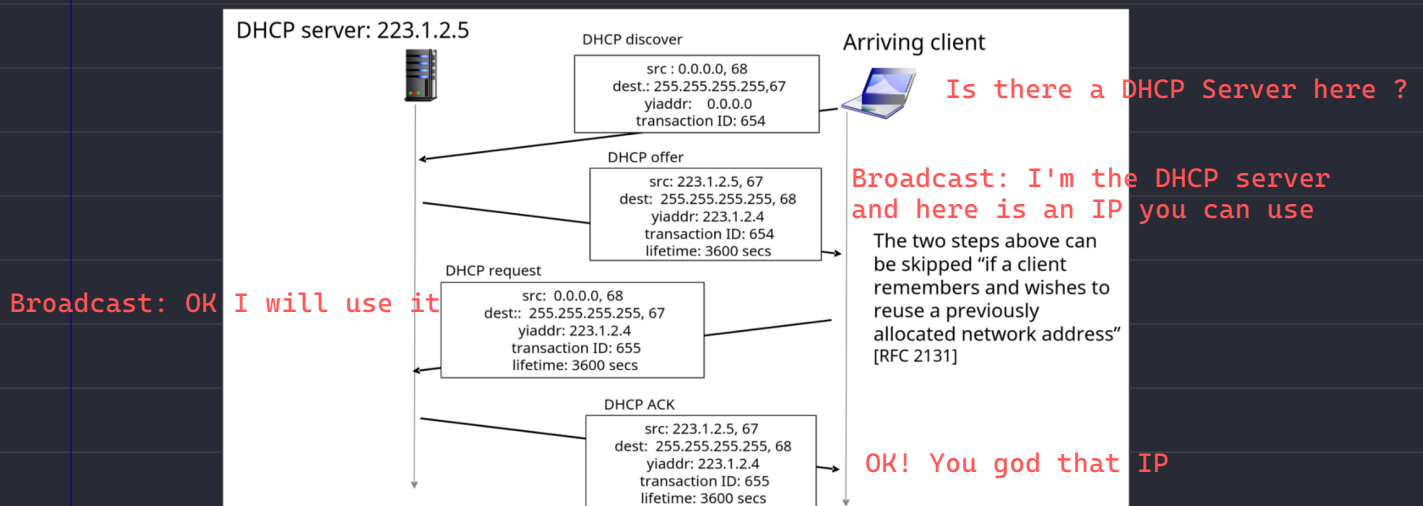
Host dynamically obtains IP address from network server when "joins" network

Can renew its lease on address in use

Allows reuse of addresses

Support for mobile users who join/leave network

DHCP client-server scenario



DHCP 使用 Application layer 通訊，並使用 UDP 連線

Generalized Forwarding, SDN

Destination-based forwarding: Forward based on dest. IP address

Generalized forwarding: many header fields can determine action

- many action possible: drop/copy/modify/log packet

Flow table abstraction

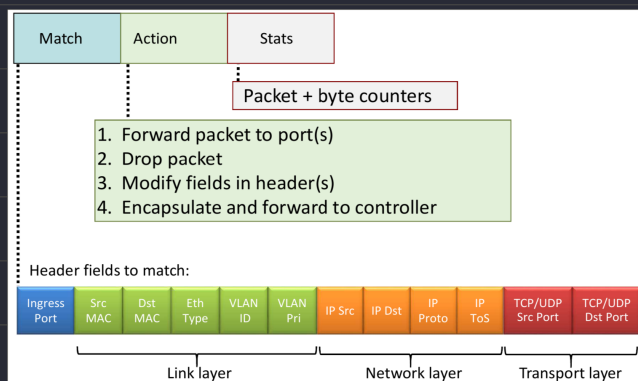
flow: defined by header fields

↗ generalized forwarding: simple packet-handling rules

- match: pattern values in packet header fields
- actions: for matched packet: drop, forward, modify, matched packet or send matched packet to controller
- priority: disambiguate overlapping patterns
- counters: #bytes and #packets

OpenFlow (Protocol): flow table entries

相較於傳統路由器只能藉由 Dst IP判斷，此協議可以根據更多層（網路層以外）的資訊決定路由行為



Examples:

Router: Match: longest destination IP prefix

Action: Forward

Switch: Match: destination MAC address

Action: forward or flood

Firewall: Match: IP addresses and TCP/UDP port numbers

Action: permit or deny

Switch Port	MAC src	MAC dst	Eth type	VLAN ID	VLAN Pri	IP Src	IP Dst	IP Prot	IP ToS	TCP s-port	TCP d-port	Action
*	*	*	*	*	*	*	*	*	*	*	22	drop

Block (do not forward) all datagrams destined to TCP port 22 (ssh port #)

NAT: Match: IP address and port

Action: rewrite address and port

“match plus action” abstraction: match bits in arriving packet header(s) in any layers, take action

- matching over many fields (link-, network-, transport-layer)
- local actions: drop, forward, modify, or send matched packet to controller

• “program” network-wide behaviors

↗ simple form of “network programmability”

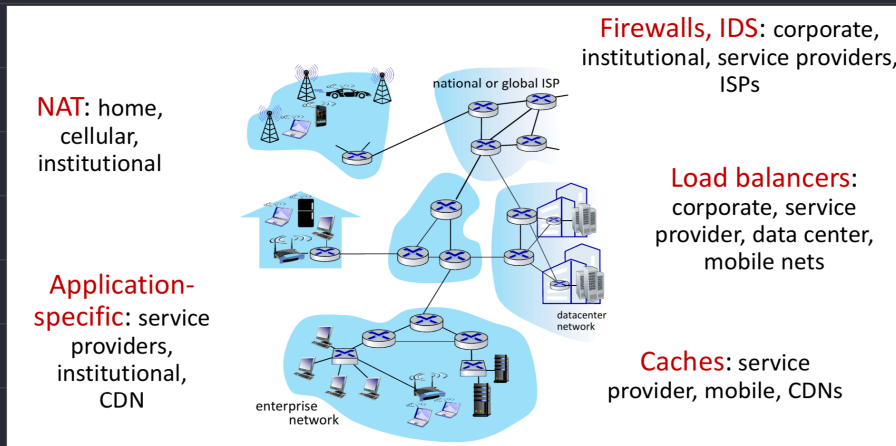
programmable per packet “processing”

Middleboxes

Definition

Any intermediary box performing functions apart from normal, standard functions of an IP router on the data path between a source host and destination host

Example



- initially: proprietary (closed) hardware solutions
- move towards “whitebox” hardware implementing open API
 - move away from proprietary hardware solutions
 - programmable local actions via match+action
 - move towards innovation/differentiation in software
- SDN: (logically) centralized control and configuration management often in private/public cloud
- network functions virtualization (NFV): programmable services over white box networking, computation, storage